

AI's not just about GPUs anymore

It has barely been two weeks since DeepSeek made a grand entrance into the artificial intelligence arena, capturing the attention of established technology organisations worldwide. Soon to be a household name, DeepSeek has caused a level of concern among companies that have long dominated this space. This reaction stems partly from DeepSeek's claims of having found a more efficient method of training and deploying AI models, alongside assertions of achieving this at a fraction of the cost borne by its competitors. Whether these cost-saving claims hold up under scrutiny remains to be seen, but the industry has already begun to speculate on how such a development could alter the broader AI landscape.

One of the rumoured reasons for DeepSeek's apparent cost-effectiveness is its alleged access to a large number of NVIDIA GPUs. However, it is worth remembering that the AI sector, as a whole, has not always been straightforward about the sources of hardware or datasets it uses. As such, it might be premature to single out DeepSeek before the facts are clarified.

This situation unfolds amidst a broader climate of apprehension about AI models originating from China. Violations of copyright laws and the questionable use of personal data in training datasets are issues that have plagued the entire industry. Consequently, the debate on whether Chinese AI firms pose a greater risk can sometimes seem like the pot calling the kettle black.

Meanwhile, much of the conversation continues to focus on performance metrics rather than the environmental or infrastructural implications of large-scale AI. The strain on existing power infrastructure is another pertinent topic. Beyond the moral and legal dimensions of hardware procurement, companies seeking to deploy advanced AI models must rely on electricity grids that may not be equipped to handle surging demand.



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A parallel question centres on whether China, in particular, might be better positioned to expand its power generation infrastructure in tandem with demand. If Chinese AI firms can leverage swift advancements in energy capacity, they might be able to train and deploy AI models without suffering from the bottlenecks that other regions could face. However, this advantage could be offset by international scrutiny and ongoing debates.

It remains to be seen whether DeepSeek's approach will genuinely revolutionise AI training practices. DeepSeek's success could well hinge on whether it can balance these factors: proving its claims of reduced expenditure, satisfying regulatory questions about hardware origins, and having reliable energy arrangements in place.

Beyond DeepSeek's individual prospects, the power-generation race has become an increasingly relevant aspect of the broader AI contest. As the performance of models steadily improves, their training requirements escalate at a corresponding rate, making efficient energy consumption strategies all the more crucial. The pace of AI development may outstrip the evolution of traditional power grids, forcing companies and governments to rethink how and where they invest in infrastructure. This realisation could spur new forms of collaboration between technology companies and energy providers, or lead to further exploration into alternative energy solutions such as renewables, hydrogen, or geothermal sources. **d**